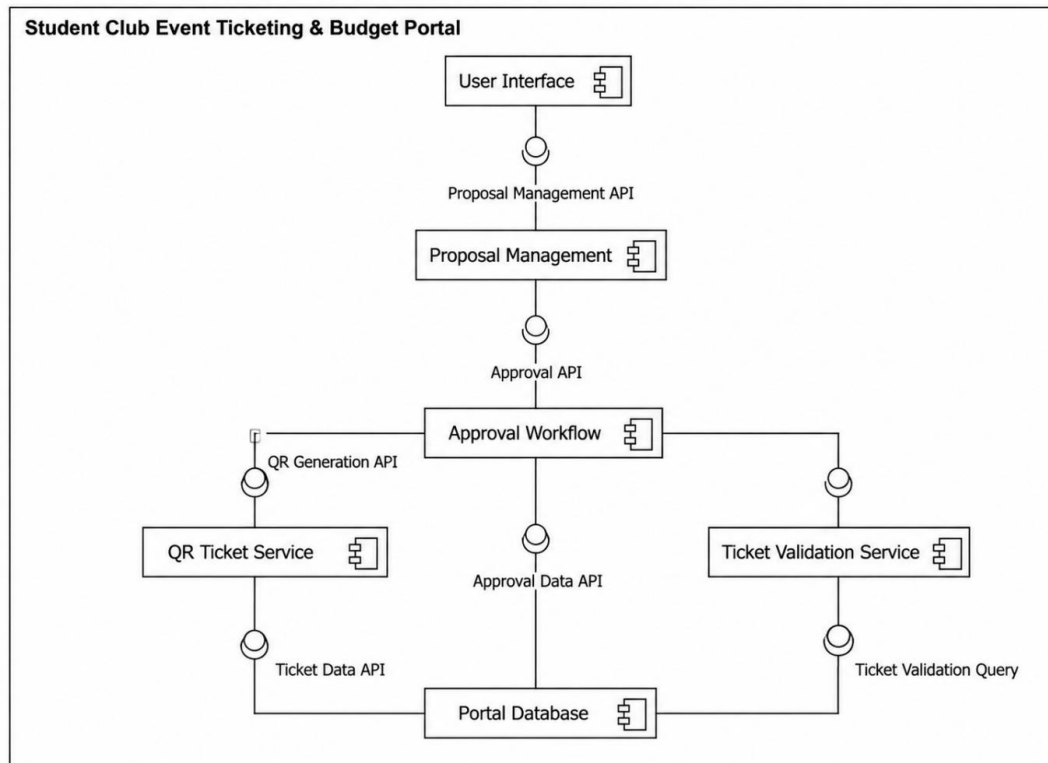


Student Club Event Ticketing & Budget Portal

SRN: PES1UG24CS378

NAME: Riya Saju Vithayathil

UML component diagram



Architecture: Layered Architecture

Architectural Choice:

The system is organized into separate layers based on their responsibilities:

- Presentation Layer: User Interface
- Business/Application Layer: Proposal Management and Approval Workflow
- Service Layer: QR Ticket Service and Ticket Validation Service
- Data Layer: Portal Database

This separation allows each part of the system to focus on a specific responsibility while communicating with the required components through defined interfaces.

Reason 1 – Multi-Stage Approval Workflow

The system requires event budget proposals to pass sequentially through the Faculty Coordinator → Finance Office → Dean. A layered structure separates the approval workflow from the user interface and data storage, making the approval process easier to manage and maintain.

Reason 2 – Separation of Ticketing Responsibilities

The system handles both QR ticket generation and ticket validation. Keeping these functions in separate service components allows ticket generation and venue validation to operate independently while interacting with the proposal and database components through defined interfaces.

Security Advantage

Layered architecture provides separation between the user interface, business logic, and database. Access to sensitive operations such as budget approval, ticket generation, and ticket validation can therefore be controlled through the appropriate application/service layers instead of allowing direct access to the database.

Performance Benefit

Separating QR generation and ticket validation into dedicated service components allows these operations to be handled independently. This can improve responsiveness during venue check-in, where tickets need to be validated quickly, and supports the requirement for timely ticket processing.